

Statistical Housing Valuation with Multiple Linear Regression

A statistical modeling and predictive analytics project focused on residential housing valuation using multiple linear regression, feature engineering, transformation techniques, and regression diagnostics in R.

1 Introduction

1.1 Research Question of Interest

This project investigates how to explain the variation in housing sale prices using a multiple linear regression (MLR) model. We also assess the model's predictive power with testing data.

1.2 Methodology

To model how housing features influence a houses's market value, we use MLR. First we clean the train dataset, engineer features, and remove trivial columns. Then we explore several candidate models, optimize the model via predictor transformation, manual variable selection, and stepwise selection to eliminate noncontributing variables. VIF scores and diagnostic plots help us detect multicollinearity and ensure the reliability of our statistical inferences.

1.3 Project Overview

This report is organized into four sections:

- **Data Cleaning:** We outline the preparation of the raw dataset, including the removal of zero-variance features, ordinal and categorical encoding, and the extraction of numerical sentiment scores from text-based reviews via natural language processing.
- **Data Description:** We provide a summary of the response variable, SalePrice, and analyzes the correlations between key structural predictors to identify significant market trends.
- **Results and Interpretation:** We detail the construction of the multiple linear regression model, covering feature engineering, power transformations, and backward stepwise selection. The section concludes with a thorough evaluation of the final model's diagnostics.
- **Discussion:** We summarize our key findings, discusses their practical implications for real estate valuation, addresses study limitations, and suggests directions for future research.

2 Data Cleaning

We processed the original dataset by neutralizing noise, standardizing categorical variables, and creating useful features. First, we removed zero-variance columns and rounded "decimal noise" in discrete fields, such as year and quality ratings, to restore their proper scales.

To improve interpretability, we converted raw dates into relative metrics, such as "House/Garage Age" and "Remodel Age," by subtracting the year of construction or remodeling from the year of sale. We standardized categorical variables by enforcing a strict hierarchy for ordinal factors ($Po < Fa < Ta < Gd < Ex$) and encoding binary variables as 1/0.

Additionally, we quantified the house review text with a simple NLP, which weights positive and negative keywords to create a numeric review sentiment score. We also created various interaction terms to better capture the home's overall features.

- **TotalArea:** A sum of all living, basement, and porch spaces

- **AboveBath, BsmtBath:** Combined full and half-baths into a single continuous metric for simplicity
- **OverallQual * TotalArea:** A square foot in a "High Quality" home is worth more than a square foot in a "Low Quality" one.

In the end, we drop the raw columns used to create new features to keep the dataset lean and avoid multicollinearity.

3 Data Description

3.1 Initial Model

We begin with an initial multiple linear regression model designed to balance interpretability and predictive power. The response variable is the logarithm of the sale price, and the model incorporates several transformed continuous variables based on PowerTransform output (Figure: 1), ordinal quality ratings, and categorical housing characteristics. These transformations address skewness and potential nonlinear relationships, while the ordinal and categorical variables capture structural and locational effects.

yjPower Transformations to Multinormality				
	Est	Power	Rounded Pwr	Wald Lwr Bnd Wald Up Bnd
Qual_x_TotalArea	0.3359		0.33	0.3195 0.3523
TotalArea	0.4652		0.47	0.4411 0.4893
GrLivArea	0.2138		0.21	0.1710 0.2565
LotArea	0.0638		0.06	0.0442 0.0833
House_Age	0.3461		0.33	0.3275 0.3647
GarageArea	0.8952		0.90	0.8766 0.9138
SalePrice	0.0656		0.07	0.0555 0.0757

Likelihood ratio test that all transformation parameters are equal to 0

Figure 1: Estimated Power Transformation Parameters for Key Variables

$$\begin{aligned}
\log(\text{SalePrice}) = & \beta_0 + \beta_1(\text{Qual_x_TotalArea})^{1/3} + \beta_2 \log(\text{GrLivArea}) + \beta_3 \log(\text{LotArea}) \\
& + \beta_4(\text{House_Age})^{1/3} + \beta_5 \text{GarageArea} + \beta_6 \log(1 + \text{BsmtFinSF1}) \\
& + \beta_7 \text{OverallQual} + \beta_8 \text{OverallCond} + \beta_9 \text{BsmtQual} \\
& + \beta_{10} \text{KitchenQual} + \beta_{11} \text{ExterQual} + \beta_{12} \text{Neighborhood} \\
& + \beta_{13} \text{MSZoning} + \beta_{14} \text{BldgType} + \beta_{15} \text{Foundation} \\
& + \beta_{16} \text{SaleCondition} + \beta_{17} \text{Functional} + \varepsilon \quad (1)
\end{aligned}$$

3.2 Summary Statistics

This correlation matrix (Figure 3: Correlation matrix of the predictors in the initial model) reveals several significant predictors of residential real estate pricing. It shows the strong relationship between **SalePrice**, **GrLivArea**, and **Qual_x.TotalArea**. The plots clearly show a positive linear correlation, suggesting that as square footage and material quality increase, prices rise predictably. In contrast, the **House_Age** variable displays a negative trend with price and quality, suggesting that newer homes generally have higher market values. The striated patterns in the **OverallQual** and **OverallCond** columns are also visible. These are ordinal rankings from 1 to 10 that act as discrete tiers, pulling the sale price upward as the rating improves.

Description: df [9 × 2]

	Mean <dbl>	SD <dbl>
SalePrice	1.219837e+06	4.523943e+06
Qual_x_TotalArea	2.076215e+04	1.103433e+04
GrLivArea	1.508931e+03	5.100154e+02
LotArea	1.034806e+04	9.919371e+03
House_Age	3.636371e+01	3.035290e+01
GarageArea	4.742861e+02	2.100834e+02
BsmtFinSF1	4.469135e+02	4.623010e+02
OverallQual	6.115429e+00	1.421003e+00
OverallCond	5.550571e+00	1.102420e+00

Figure 2: Summary statistics of main continuous variables.

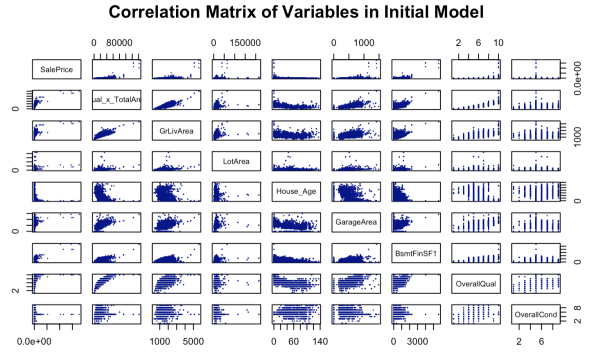


Figure 3: Correlation matrix of the predictors in the initial model.

We first examine the distribution of the response variable **SalePrice** using a histogram (Figure 4: Histogram of SalePrice). The distribution is highly right-skewed, with a long tail toward very large house prices, suggesting that a few extreme values may dominate the scale and violate the normality assumption of linear regression. To address this issue, we apply a log transformation. After taking $\log(\text{SalePrice})$, the distribution becomes more symmetric and approximately bell-shaped. This transformation reduces skewness and stabilizes the variance, making the data more suitable for linear regression. Therefore, we use $\log(\text{SalePrice})$ as the response variable in subsequent analysis.

Distribution of Continuous Variables (See Appendix Figure 6) The continuous variables generally show substantial right-skewness and several spikes at zero, indicating that many housing-related size measures are nonnegative, concentrated among smaller values, and include some homes with missing or absent features. A few variables also show long right tails or mild multimodality, suggesting the presence of outliers and heterogeneous housing stock.

Distribution of Ordinal Variables (See Appendix Figure 7) The ordinal variables are concentrated in the middle-to-upper categories, indicating that most homes in the dataset are of average to good quality and condition rather than at the extremes. Overall, these distributions suggest moderate variation, with quality showing somewhat more spread than condition.

Distribution of Categorical Variables (See Appendix Figure 8) The categorical variables are unevenly distributed, with several categories dominating while others appear only rarely, indicating noticeable class imbalance across housing characteristics. At the same time, variables such as neighborhood and sale-related categories still show meaningful heterogeneity, suggesting that location and market context may help explain price variation.

4 Results and Interpretations

4.1 R Results

Our model demonstrates exceptional predictive power, as evidenced by an adjusted R^2 of 0.9346. This indicates that 93.46% of the variance in house prices is explained by the included predictors, even after adjusting for model complexity. This high explanatory power is supported by an F -statistic of 1,318, which confirms the model's overall significance with a p -value near zero ($p < 2.2 \times 10^{-16}$). Structurally, TotalArea emerges as the most influential driver, showing strong positive elasticity: larger properties command significantly higher prices. Qualitative metrics, such as OverallQual and engineered ReviewSentiment scores, suggest that material luxury and qualitative market descriptors provide substantial value premiums. In contrast, negative coefficients for House_Age and specific foundations, such as Slab, highlight clear market depreciators. Notably, sev-

eral predictors with p -values above the 0.05 threshold were retained, including specific levels within Neighborhood, because they are crucial for controlling for spatial heterogeneity and ensuring the model captures localized market nuances that would otherwise bias the remaining estimates.

4.2 Candidate Models

We considered three candidate multiple linear regression models during the model-building process.

Model 1 (Initial Model): The first model uses `SalePrice` as the response and includes several key housing characteristics, such as `Qual_x_TotalArea`, `TotalArea`, `OverallQual`, `GrLivArea`, `House_Age`, `Neighborhood`, `OverallCond`, `GarageArea`, `LotArea`, and `KitchenQual`. This model serves as a baseline specification using the original response scale and untransformed predictors.

Model 2 (Refined Initial Model): The second model replaces `SalePrice` with $\log(\text{SalePrice})$ and applies transformations to several skewed continuous predictors, including cube-root transformations for `Qual_x_TotalArea` and `House_Age`, and log transformations for `GrLivArea` and `LotArea`. This model retains the most important variables from the initial specification while improving linearity and variance stability.

Model 3 (Improved Model): The third model begins with Model 2, removes predictors exhibiting substantial multicollinearity based on VIF (See Appendix Figure 9), and then introduces additional candidate variables to capture other potentially important housing characteristics. We then applied stepwise regression based on the AIC criterion to select a more parsimonious specification, and finally further reduced the resulting AIC-selected model to 20 variables by minimizing RMSE. In addition, for transformed continuous predictors, we used log transformations rather than power transformations when possible in order to improve interpretability. This specification is designed to better capture structural, locational, and market-related variation in housing prices while maintaining transformed predictors for improved model fit.

4.3 Final Model

$$\begin{aligned} \log(\text{SalePrice}) = & \beta_0 + \beta_1 \log(1 + \text{TotalArea}) + \beta_2 \text{ReviewSentiment} + \beta_3 \text{Foundation} \\ & + \beta_4 \text{HouseStyle} + \beta_5 \text{OverallQual} + \beta_6 \log(1 + \text{BsmtUnfSF}) \\ & + \beta_7 \log(1 + \text{BsmtFinSF1}) + \beta_8 \text{BsmtFinType1} + \beta_9 \log(1 + \text{GarageArea}) \\ & + \beta_{10} \log(1 + \text{OutdoorArea}) + \beta_{11} \text{Neighborhood} + \beta_{12} \text{AboveBath} \\ & + \beta_{13} \log(1 + \text{BsmtFinSF2}) + \beta_{14} \log(1 + \text{House_Age}) + \beta_{15} \log(1 + \text{LotArea}) \\ & + \beta_{16} \text{Functional} + \beta_{17} \text{PoolQC} + \beta_{18} \text{GarageType} \\ & + \beta_{19} \text{RoofMat1} + \beta_{20} \text{TotRmsAbvGrd} + \varepsilon \quad (2) \end{aligned}$$

This final model produces a Kaggle score of 859683.28753.

4.3.1 Interpretation (Response Variable: $\log(\text{SalePrice})$)

1. **Log-transformed continuous variables:** For `TotalArea`, `BsmtUnfSF`, `BsmtFinSF1`, `GarageArea`, `OutdoorArea`, `BsmtFinSF2`, `House_Age`, and `LotArea`, each coefficient is interpreted as an elasticity. A 1% increase in the predictor is associated with an approximate $\beta_j\%$ change in `SalePrice`, holding all other variables constant.

2. **Integer count-based variables:** For OverallQual, AboveBath, and TotRmsAbvGr, a one-unit increase in the predictor is associated with an approximate $100 \times \beta_j \%$ change in SalePrice, holding all other variables constant. The exact effect is $(e^{\beta_j} - 1) \times 100\%$.
3. **Categorical variables with dummy coding:** For ReviewSentiment, Foundation, HouseStyle, BsmtFinType1, Neighborhood, Functional, PoolQC, GarageType, and RoofMat1, each dummy coefficient measures how the predicted SalePrice for one category compares to the baseline category, holding all other variables constant. In particular, if the coefficient for a category is β_j , then houses in that category are predicted to have $(e^{\beta_j} - 1) \times 100\%$ higher or lower SalePrice than houses in the baseline category.

4.3.2 Justification

After evaluating several candidate models, we selected the above specification as the final predictive model because it achieves the best balance between predictive performance and model simplicity. While earlier models included a larger set of predictors, the final model retains only variables that meaningfully contribute to explaining variation in housing prices, thereby avoiding unnecessary complexity. Key predictors capturing housing size, structural characteristics, quality, and location are preserved, and appropriate transformations are applied to reduce skewness and improve the linear relationships with the response variable. These choices improve model stability and predictive accuracy. Compared with the previous candidate models, this specification provides strong predictive performance with a lower RMSE while maintaining interpretability. Therefore, we consider this model to be the most appropriate and reliable specification for predicting housing prices.

4.4 Model Assessment

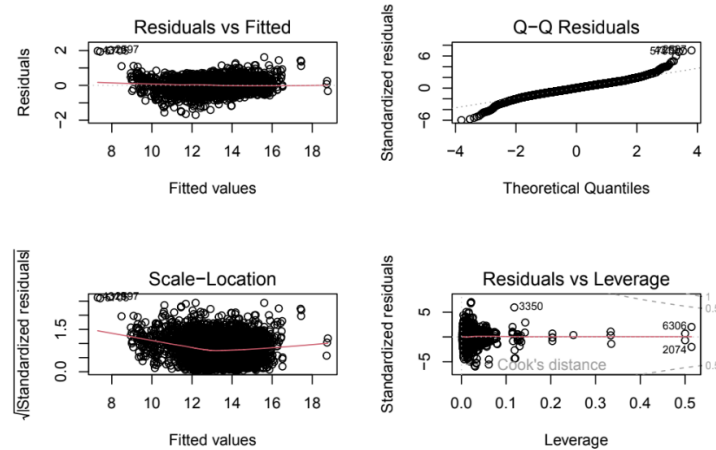


Figure 4: Diagnostic plots of the final model.

We assessed the final model using several diagnostic tools (5 4)

- The Residuals vs. Fitted plot shows that the residuals are generally randomly scattered around the horizontal line at zero. There is no obvious patterns, suggesting that the relationship between the predictors and the response is linear.
- The Normal Q-Q plot shows that most of the points follow the straight line, suggesting that the error terms are approximately normally distributed. We observe some "heavy tails" at the extreme ends. However, for a dataset of this size, the overall normality is reasonable enough.

- The Scale-Location plot shows a relatively random standardized residuals . While the red smoothing line shows a slight dip, it remains relatively flat overall. Thus the assumption of constant variance is satisfied.
- The Residuals vs. Leverage plot shows that all observations fall within the boundaries of Cook's Distance. Although points like 3350 and 6306 have higher leverage, they do not have high residuals, meaning they would not negatively alter our regression results if removed.

Conclusion: The diagnostic plots indicate that the final model is valid and reliable.

5 Discussion

5.1 Summary

This project examines the extent to which housing characteristics explain variation in residential sale prices using a multiple linear regression framework. Beginning with the raw housing dataset, we conducted systematic data cleaning, feature engineering, and careful encoding of categorical and ordinal variables to construct meaningful predictors that capture structural, quality, and neighborhood-related attributes. Exploratory analysis revealed substantial skewness in several continuous variables, motivating the use of log and power transformations to improve linearity and variance stability. We then developed and compared multiple candidate models, progressively refining the specification to balance predictive accuracy, interpretability, and model parsimony. The resulting final model integrates key determinants of housing value, including property size, construction quality, structural features, and neighborhood effects. Diagnostic analyses indicate that the model satisfies the core assumptions of linear regression and provides a robust and interpretable framework for explaining variation in housing prices.

5.2 Real-World Implications

The results of our final model are consistent with well-established findings in the housing economics literature. According to the hedonic pricing framework proposed by Rosen (1974), housing prices reflect the combined value of a bundle of property characteristics such as size, structural quality, and neighborhood attributes. Our model identifies similar determinants, as variables capturing housing size, quality, and location play significant roles in explaining variation in sale prices. Empirical studies have also shown that structural housing attributes strongly influence property values (Goodman, 1988). In addition, the importance of neighborhood variables in our model is supported by spatial housing price research demonstrating that neighborhood quality and location significantly affect housing prices (Dubin, 1992). These findings suggest that the final model is consistent with real-world housing market dynamics.

5.3 Limitation

While our final model demonstrates high predictive accuracy and overall validity, several limitations regarding its statistical validity and structural simplicity remain:

- Complexity vs. Simplicity: The model relies on 20 terms, but many are categorical factors with numerous levels. For example, Neighborhood alone introduces 24 separate coefficients, making the model harder to interpret for non-statisticians.
- Categorical Reduction: We have not yet identified a way to group these categories (e.g., clustering neighborhoods) without significantly reducing the model's predictive capacity.

Future Work: We can work out a more refined level collapsing logic of categorical variables to achieve higher simplicity. Also, although this was not covered in the content of this class, using LASSO regression might be able to force the less impactful categorical levels to zero.

6 Works Cited

References

- [1] Dubin, Robin A. “Spatial Autocorrelation and Neighborhood Quality.” *Regional Science and Urban Economics*, vol. 22, no. 3, 1992, pp. 433–452.
- [2] Goodman, Allen C. “An Econometric Model of Housing Price, Permanent Income, Tenure Choice, and Housing Demand.” *Journal of Urban Economics*, vol. 23, no. 3, 1988, pp. 327–353.
- [3] Rosen, Sherwin. “Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition.” *Journal of Political Economy*, vol. 82, no. 1, 1974, pp. 34–55.

7 Appendix

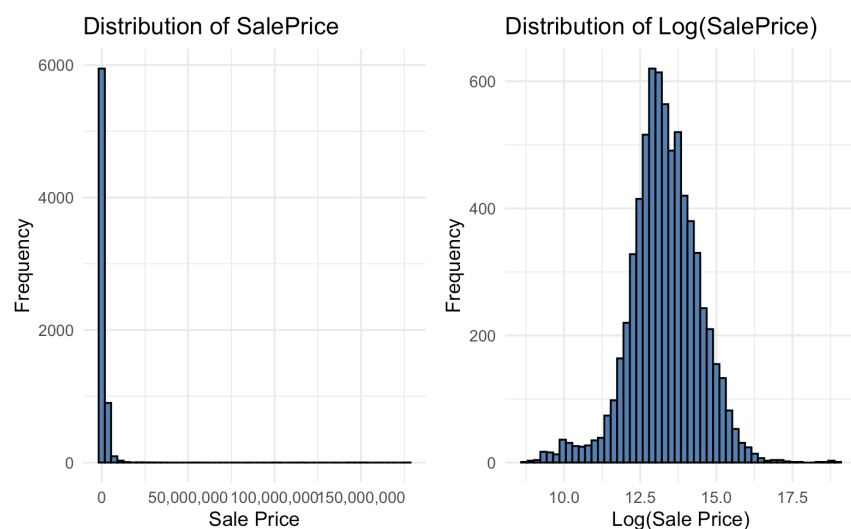


Figure 5: Histogram of SalePrice

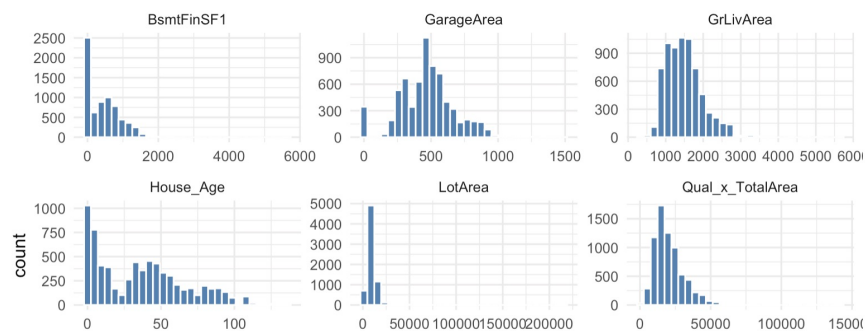


Figure 6: Distribution of Continuous Variables

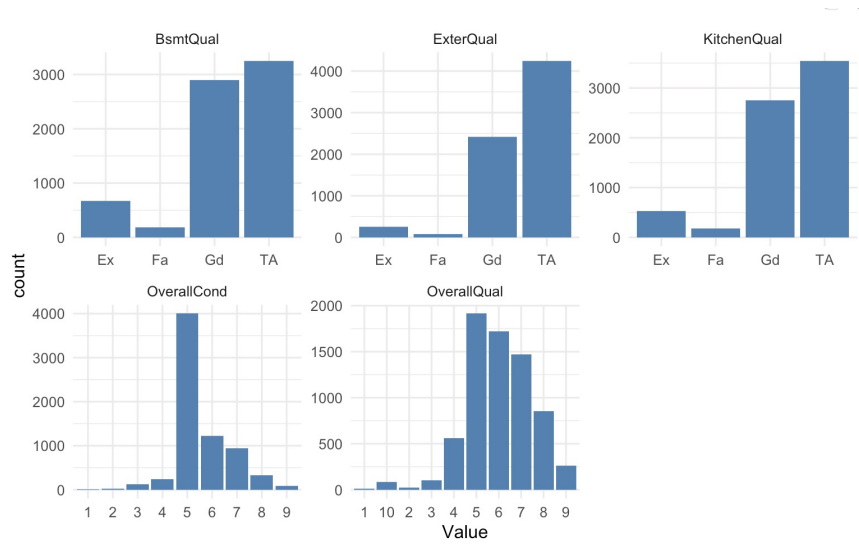


Figure 7: Distribution of Ordinal Variables

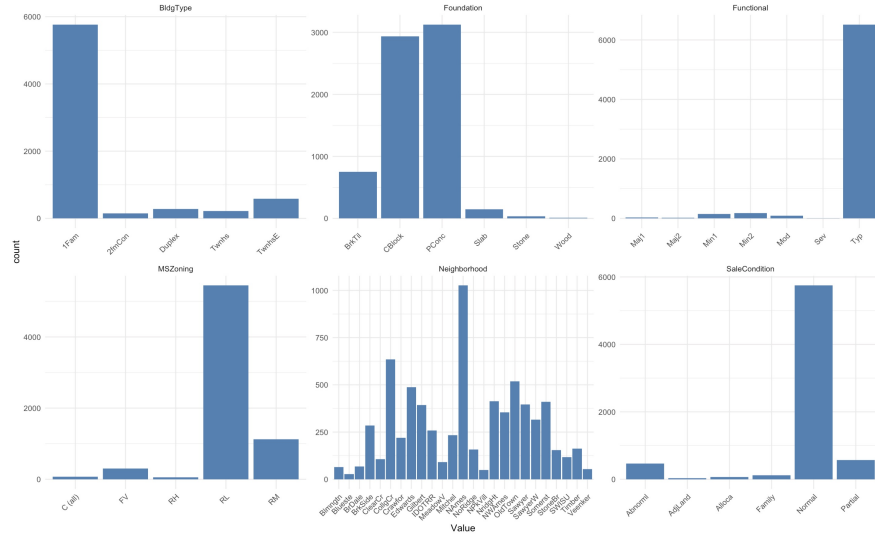


Figure 8: Distribution of Categorical Variables

##	GVIF	Df	GVIF ^{1/(2*Df)}
## Qual_x_Area	54.46701	1	7.380177
## Garage_x_Qual	104.95184	1	10.244601
## GarageCars_cap4	55.58169	1	7.455313
## Qual_x_TotalArea	117.34501	1	10.832590
## Cond_x_TotalArea	50.06917	1	7.075957
## House_Age_x_TotalArea	40.85734	1	6.391974
## log1p(House_Age)	60.73604	1	7.793333
## log1p(TotalArea)	34.37842	1	5.863311

Figure 9: Transformed Model VIF


```
##
## Call:
## lm(formula = Log_SalePrice ~ logip(TotalArea) + ReviewSentiment +
##   Foundation + HouseStyle + OverallQual + logip(BsmtUnfSF) +
##     logip(BsmtFinSF1) + BsmtFinType1 + logip(GarageArea) + logip(OutdoorArea) +
##   Neighborhood + AboveBath + logip(BsmtFinSF2) + logip(House_Age) +
##     logip(LotArea) + Functional + PoolQC + GarageType + RoofMatl +
##   TotRmsAbvGrd, data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.69582 -0.17488  0.00293  0.18169  2.00884
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -2.111394    0.367689  -5.742 9.73e-09 ***
## logip(TotalArea)  1.794051    0.028116  63.808 < 2e-16 ***
## ReviewSentiment-1  0.577299    0.013767  41.934 < 2e-16 ***
## ReviewSentiment0  0.567947    0.008910  63.741 < 2e-16 ***
## ReviewSentiment1  1.167388    0.017480  66.782 < 2e-16 ***
## ReviewSentiment2  1.144228    0.010497  109.001 < 2e-16 ***
## ReviewSentiment3  1.166219    0.017166  67.937 < 2e-16 ***
## FoundationCBlock -0.033314    0.015161  -2.197 0.028031 *
## FoundationPConc  -0.037914    0.017186  -2.206 0.027411 *
## FoundationSlab    -0.746401    0.037935 -19.676 < 2e-16 ***
## FoundationStone  -0.156852    0.051355  -3.054 0.002265 **
## FoundationWood    -0.097130    0.106474  -0.912 0.361670
## HouseStyle1.5Unf  0.269261    0.046268   5.820 6.16e-09 ***
## HouseStyle1Story  0.224911    0.013949  16.123 < 2e-16 ***
## HouseStyle2.5Fin  -0.383951    0.065776  -5.837 5.55e-09 ***
## HouseStyle2.5Unf  -0.051048    0.040488  -1.261 0.207419
## HouseStyle2Story  -0.113556    0.015016  -7.562 4.47e-14 ***
## HouseStyleSFoyer  0.337702    0.026635  12.679 < 2e-16 ***
## HouseStyleSLvl    0.007376    0.022255   0.331 0.740329
## OverallQual       0.179968    0.004733  38.023 < 2e-16 ***
## logip(BsmtUnfSF)  0.075390    0.003042  24.780 < 2e-16 ***
## logip(BsmtFinSF1) 0.102362    0.003854  26.563 < 2e-16 ***
## BsmtFinType1LwQ   -0.336610    0.026478 -12.713 < 2e-16 ***
## BsmtFinType1Rec   -0.420474    0.025003 -16.817 < 2e-16 ***
## BsmtFinType1BLQ   -0.440308    0.025451 -17.300 < 2e-16 ***

## BsmtFinType1ALQ   -0.436860    0.025008 -17.469 < 2e-16 ***
## BsmtFinType1GLQ   -0.469007    0.023151 -20.259 < 2e-16 ***
## logip(GarageArea) -0.070094    0.003682 -19.038 < 2e-16 ***
## logip(OutdoorArea) -0.041737    0.002331 -17.903 < 2e-16 ***
## NeighborhoodBlueste -0.026260    0.067533  -0.389 0.697402
## NeighborhoodBrDale  0.087315    0.054578   1.600 0.109684
## NeighborhoodBrkSide  0.018888    0.045385   0.416 0.677293
## NeighborhoodClearCr -0.006752    0.050916  -0.133 0.894509
## NeighborhoodCollCr  -0.016538    0.039644  -0.417 0.676563
## NeighborhoodCrawfor -0.007121    0.044804  -0.159 0.873721
## NeighborhoodEdwards  0.036301    0.042588   0.852 0.394028
## NeighborhoodGilbert  0.036648    0.041832   0.876 0.381013
## NeighborhoodIDOTRR  -0.003610    0.045640  -0.079 0.936959
## NeighborhoodMeadowV -0.004362    0.050829  -0.086 0.931614
## NeighborhoodMitchel -0.024940    0.044054  -0.566 0.571330
## NeighborhoodNames   0.032994    0.041525   0.795 0.426889
## NeighborhoodNoRidge  0.148257    0.045867   3.232 0.001234 **
## NeighborhoodNPKVill  0.181439    0.057588   3.151 0.001636 **
## NeighborhoodNridgHt  0.114992    0.040416   2.845 0.004451 **
## NeighborhoodNWames  0.080176    0.043047   1.863 0.062570 .
## NeighborhoodOldTown  0.083066    0.043787   1.897 0.057865 .
## NeighborhoodSawyer  0.139117    0.043157   3.223 0.001272 **
## NeighborhoodSawyerW  0.099222    0.041811   2.373 0.017665 *
## NeighborhoodSomerst  0.085556    0.039893   2.145 0.032019 *
## NeighborhoodStoneBr  0.161249    0.044276   3.642 0.000273 ***
## NeighborhoodSWISU   0.169140    0.050467   3.351 0.000808 ***
## NeighborhoodTimber  0.171274    0.044974   3.808 0.000141 ***
## NeighborhoodVeenker  0.082683    0.056747   1.457 0.145153
## AboveBath          -0.118455    0.009971 -11.880 < 2e-16 ***
## logip(BsmtFinSF2)  0.029492    0.002238  13.178 < 2e-16 ***
## logip(House_Age)   -0.065009    0.006145 -10.580 < 2e-16 ***
## logip(LotArea)     0.126389    0.010351  12.210 < 2e-16 ***
## FunctionalMaj2     -0.070687    0.156364  -0.452 0.651234
## FunctionalMaj1     0.234469    0.150705   1.556 0.119798
## FunctionalMod       0.178575    0.145727   1.225 0.220464
## FunctionalMin2     0.145428    0.144062   1.009 0.312780
## FunctionalMin1     -0.015130    0.143746  -0.105 0.916176
## FunctionalTyp       0.204736    0.142315   1.439 0.150306
## PoolQCgd           -0.392215    0.207913  -1.886 0.059278 .
## PoolQCEx           -0.693287    0.098187  -7.061 1.82e-12 ***
## GarageTypeAttchd   0.152972    0.037706   4.057 5.02e-05 ***
## GarageTypeBasment  0.049762    0.047631   1.045 0.296179
## GarageTypeBuiltIn  0.070162    0.041140   1.705 0.088158 .
## GarageTypeCarPort  0.240154    0.069893   3.436 0.000594 ***
## GarageTypeDetchd   0.155724    0.038415   4.054 5.10e-05 ***

## RoofMatlTar&Grv    -1.790727    0.267967  -6.683 2.53e-11 ***
## RoofMatlWdShake    -1.639826    0.275916  -5.943 2.93e-09 ***
## RoofMatlWdShngl    -1.718222    0.278264  -6.175 7.00e-10 ***
## TotRmsAbvGrd       -0.024799    0.003922  -6.323 2.73e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2887 on 6923 degrees of freedom
## Multiple R-squared:  0.9353, Adjusted R-squared:  0.9346
## F-statistic: 1318 on 76 and 6923 DF, p-value: < 2.2e-16
```

Figure 10: R Results of Final Model